3.4.2 DNA and protein synthesis

Content	Opportunities for skills development
The concept of the genome as the complete set of genes in a cell and of the proteome as the full range of proteins that a cell is able to produce. The structure of molecules of messenger RNA (mRNA) and of transfer RNA (tRNA). Transcription as the production of mRNA from DNA. The role of RNA polymerase in joining mRNA nucleotides. • In prokaryotes, transcription results directly in the production of mRNA from DNA. • In eukaryotes, transcription results in the production of pre-mRNA; this is then spliced to form mRNA. Translation as the production of polypeptides from the sequence of codons carried by mRNA. The roles of ribosomes, tRNA and ATP. **Students should be able to:** • relate the base sequence of nucleic acids to the amino acid sequence of polypeptides, when provided with suitable data about the genetic code • interpret data from experimental work investigating the role of nucleic acids. Students will **not** be required to recall in written papers specific codons and the amino acids for which they code.	